CASE STUDY: EXPLORATORY DATA ANALYSIS AND VISUALIZATION

Wisconsin Diagnostic Breast Cancer Dataset

Course: Exploratory Data Analysis and Visualization (U21ADP05)

Assignment: Assignment II | Academic Year 2026–2027

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Educational case study only — not a clinical diagnostic system.

Abstract

This case study applies exploratory data analysis (EDA), statistical summaries, visualization and deep learning to the Wisconsin Diagnostic Breast Cancer dataset. It contains 569 observations, 30 numerical input features and a binary diagnosis target. The workflow checks data quality, examines class balance and feature distributions, studies selected feature relationships, and creates five EDA visualizations. A Multilayer Perceptron (MLP) classifier is trained after a stratified train/validation/test split, with standardization fitted only on training data to reduce data leakage. Evaluation uses accuracy, malignant-class precision, recall, F1-score, a confusion matrix and ROC-AUC. On the held-out test partition, the model achieved 94.2% accuracy and 0.992 ROC-AUC for the malignant class. These results are educational and dataset-specific; they do not establish clinical reliability or suitability for patient care.

1. Introduction and Objectives

Breast cancer diagnosis involves distinguishing malignant from benign findings. This case study uses a commonly used diagnostic benchmark dataset to demonstrate how EDA can reveal structure in numerical measurements and how a neural network can learn a classification boundary. This is a data-analysis exercise, not medical advice or a clinical validation study.

Objectives: inspect dataset structure and data quality; summarize feature distributions; investigate relationships among features and diagnosis; train and evaluate a deep-learning classifier using a leakage-aware workflow; and communicate findings, limitations and reproducible instructions.

2. Dataset Description

The included CSV contains **569 rows**, **30 numerical input features** and one target column. Features summarize characteristics of cell nuclei measured from digitized fine needle aspirate images. Target encoding: 0 = malignant and 1 = benign. There are 212 malignant cases and 357 benign cases. The CSV is included at *data/breast_cancer_wisconsin.csv*.

Dataset references: scikit-learn's load_breast_cancer documentation and the UCI Machine Learning Repository entry for Wisconsin Diagnostic Breast Cancer. Full links appear in References.

3. EDA and Preprocessing

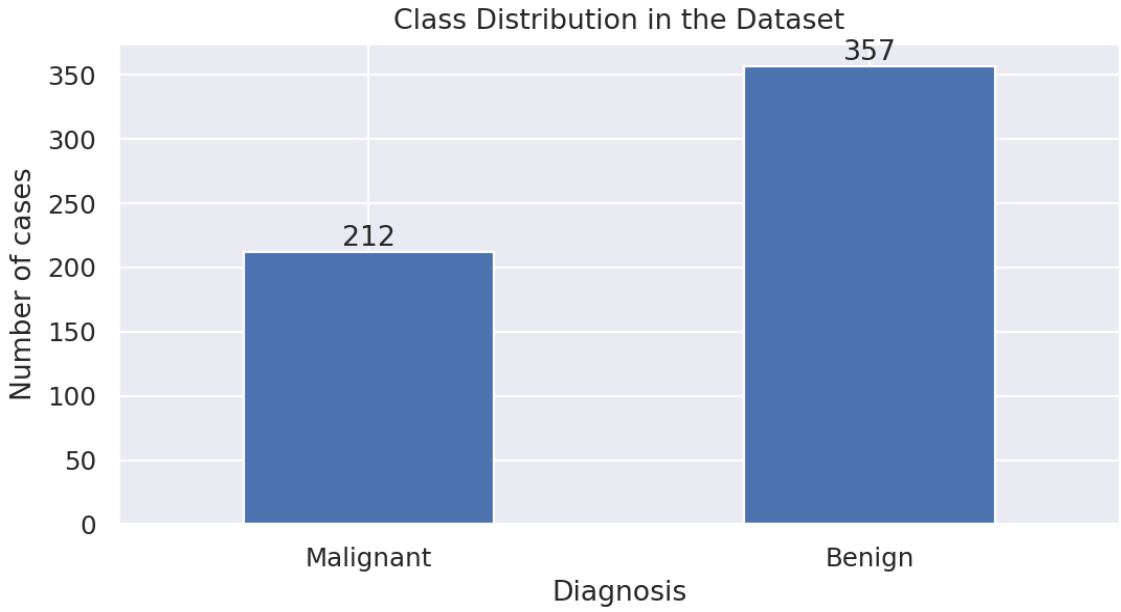
The data-quality audit found 0 missing values and 0 exact duplicate rows in the included CSV. No imputation or duplicate-row removal was required. Numerical features were summarized; selected variables were explored with distribution plots, a boxplot, a correlation heatmap and a scatterplot. Potential outliers were inspected rather than automatically removed because extreme measurements can carry useful information.

Stratified sampling created training (398 rows), validation (85 rows) and test (86 rows) partitions, approximately 70%/15%/15%. StandardScaler was fitted only on the training partition, then applied to validation and test data to avoid leakage.

4. Data Visualization

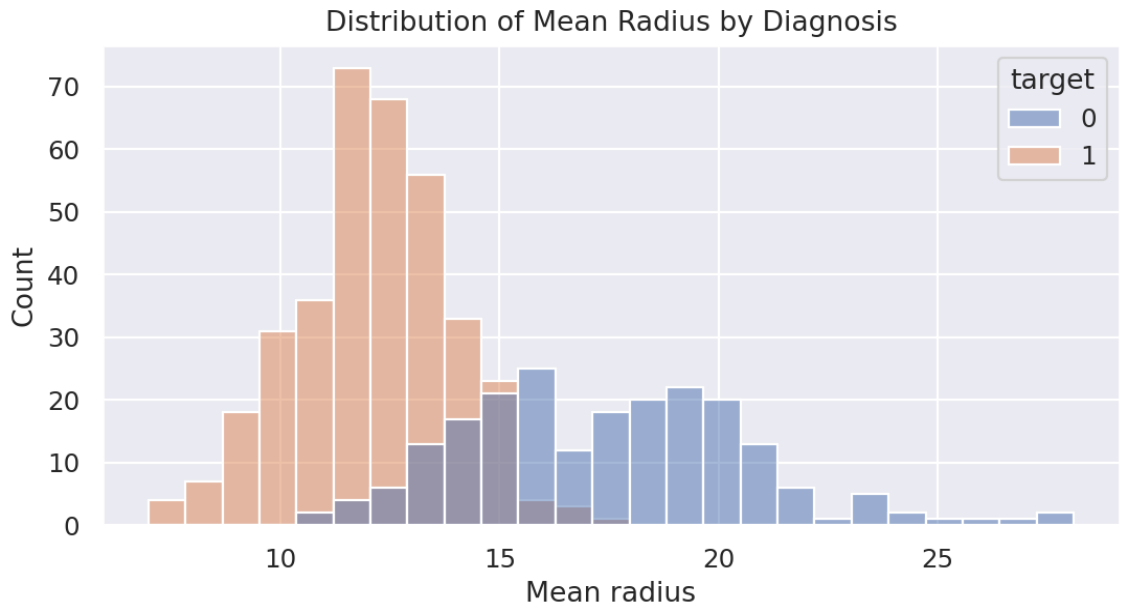
Each visualization answers a distinct exploratory question. The interpretations describe the patterns visible in the generated charts and should be rechecked if the data or random seed changes.

Figure 1. Class distribution



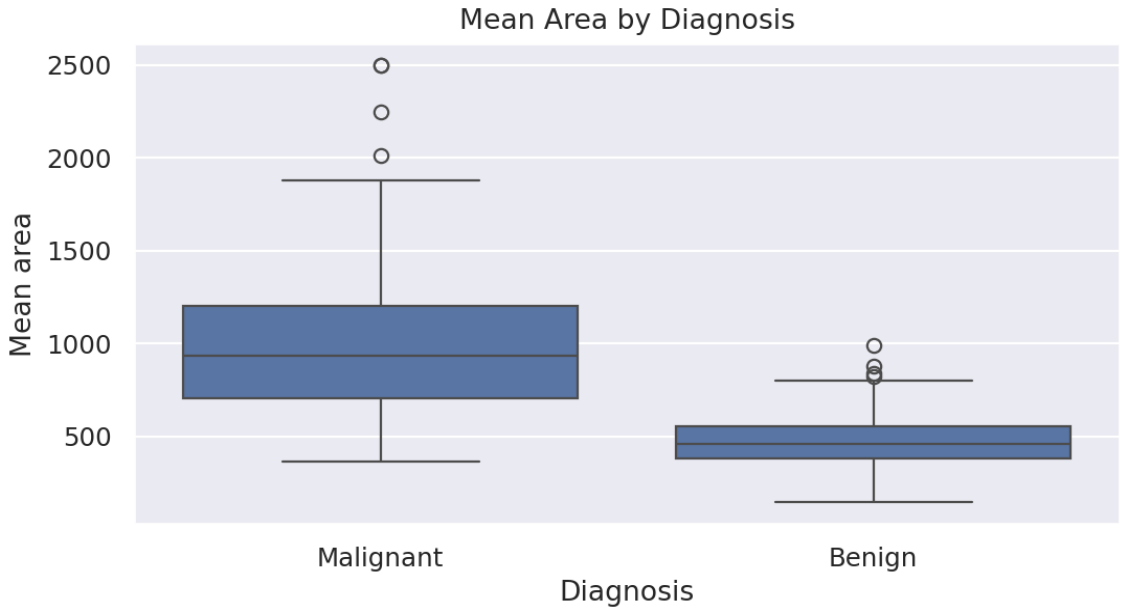
The chart compares the number of malignant and benign records. The dataset contains 357 benign and 212 malignant cases, so the classes are not perfectly balanced. Stratified splitting helps preserve a similar class proportion across partitions.

Figure 2. Mean radius distribution



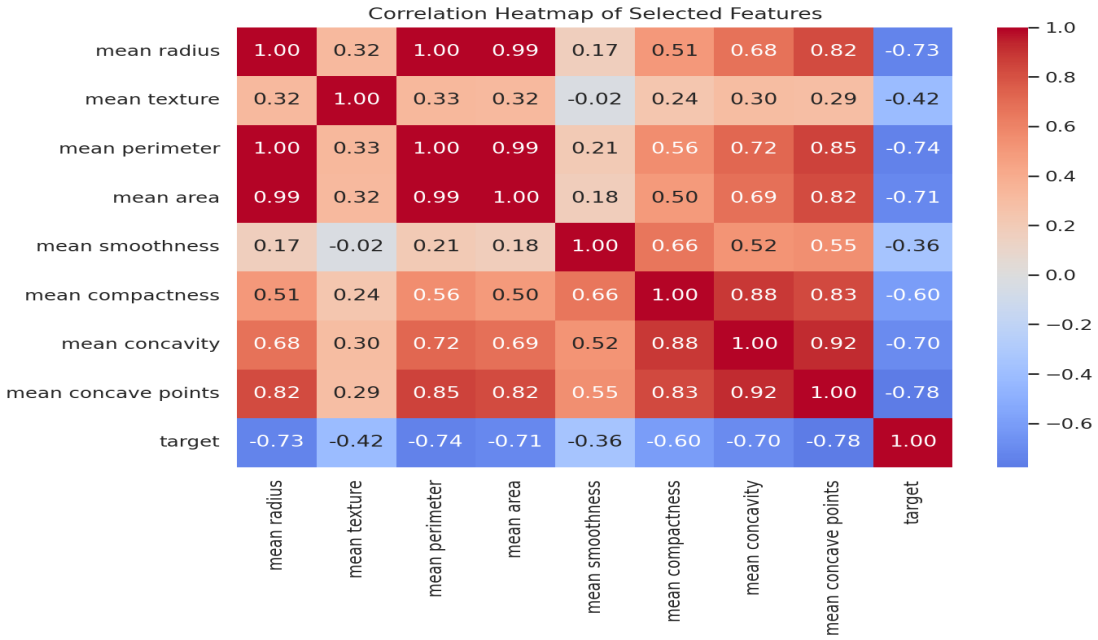
The histograms compare mean-radius values by target class. The distributions show class-related separation, although overlap remains; therefore mean radius alone should not be treated as a perfect classifier.

Figure 3. Mean area by diagnosis



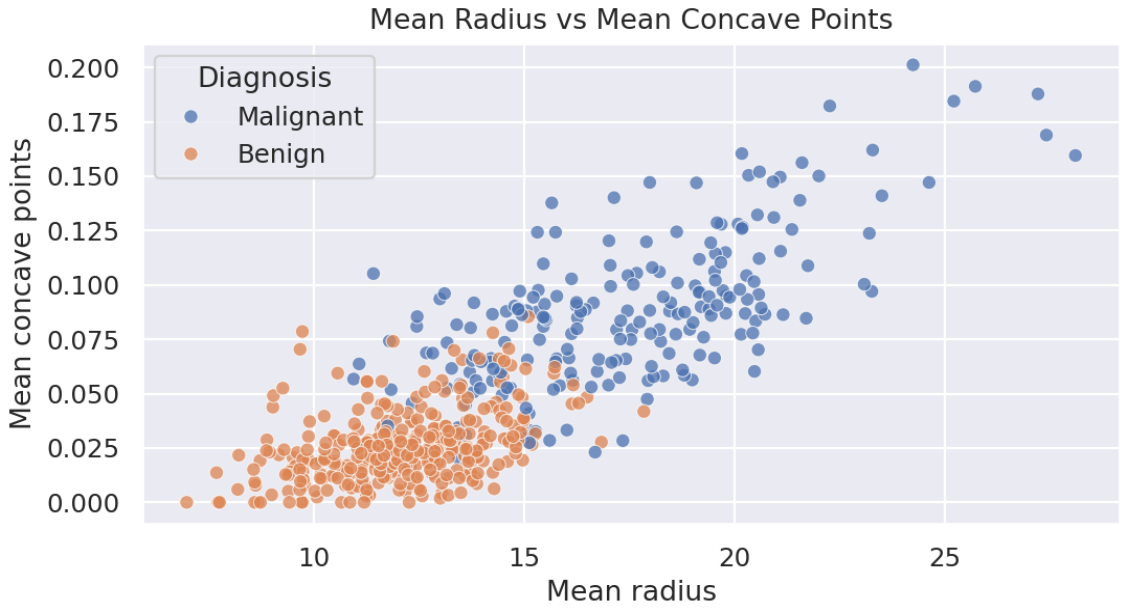
The boxplot compares the median, interquartile range and potential outliers of mean area across diagnosis groups. A shift in the distributions suggests mean area may help distinguish classes; the plot also highlights within-class variation.

Figure 4. Correlation heatmap



The heatmap displays pairwise linear correlations among selected numerical features and the encoded target. Radius, perimeter and area measurements are strongly related because they describe related aspects of size. Correlation does not establish causation, and target correlations depend on the chosen numeric encoding.

Figure 5. Feature relationship



The scatterplot compares mean radius with mean concave points and colors observations by diagnosis. Class clustering and overlap illustrate why combining multiple features can be more useful than relying on a single measurement.

5. Deep-Learning Model

A Multilayer Perceptron (MLP) was selected for structured numerical features. It uses hidden layers of 64 and 32 units, ReLU activation, Adam optimization, L2 regularization (alpha = 0.001), learning rate 0.001, early stopping and a maximum of 500 iterations. Training stopped after 51 iterations in this run. The scaler was fitted on training data only. Early stopping uses an internal validation fraction from training data; the separate validation partition provides an additional check, and the test partition is held out for final metrics.

6. Model Evaluation and Results

Separate validation accuracy: 0.988
Test accuracy: 0.942
Malignant-class precision: 0.966
Malignant-class recall: 0.875
Malignant-class F1-score: 0.918
Malignant-class ROC-AUC: 0.992

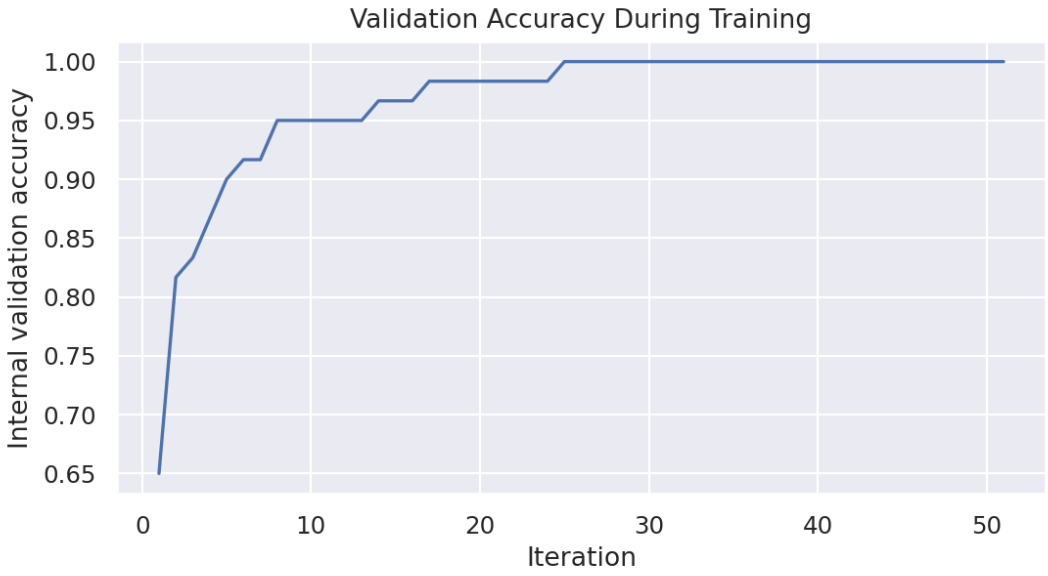
Malignant-class recall helps quantify false negatives in this benchmark: lower recall means some malignant examples were predicted as benign. These metrics must be interpreted with the small fixed test set and original data-collection context in mind.

Figure 6. Training loss



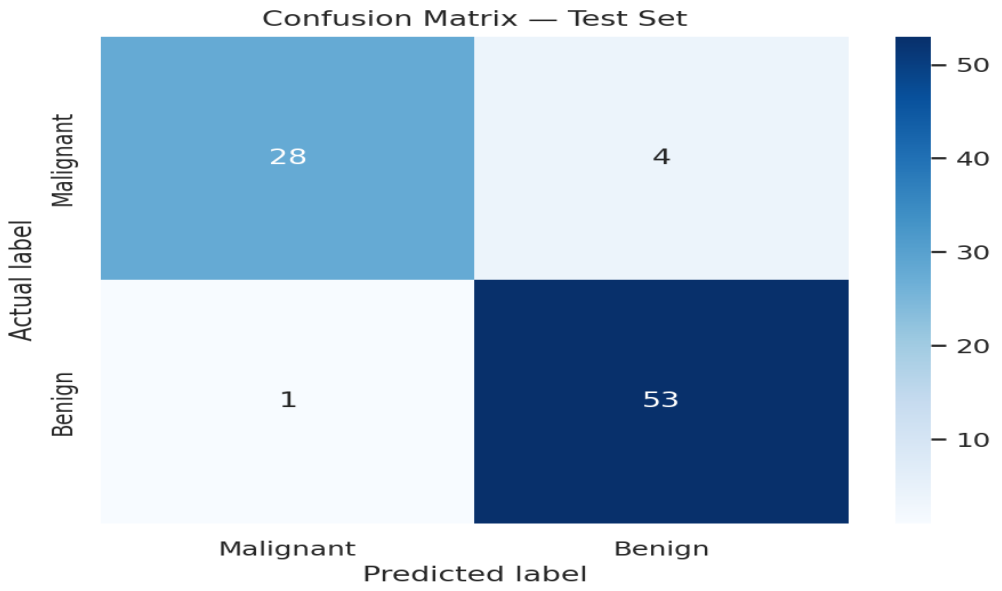
Training loss summarizes the objective optimized by the MLP. A downward trend indicates learning; the curve should be reviewed for instability or failure to converge.

Figure 7. Validation accuracy during training



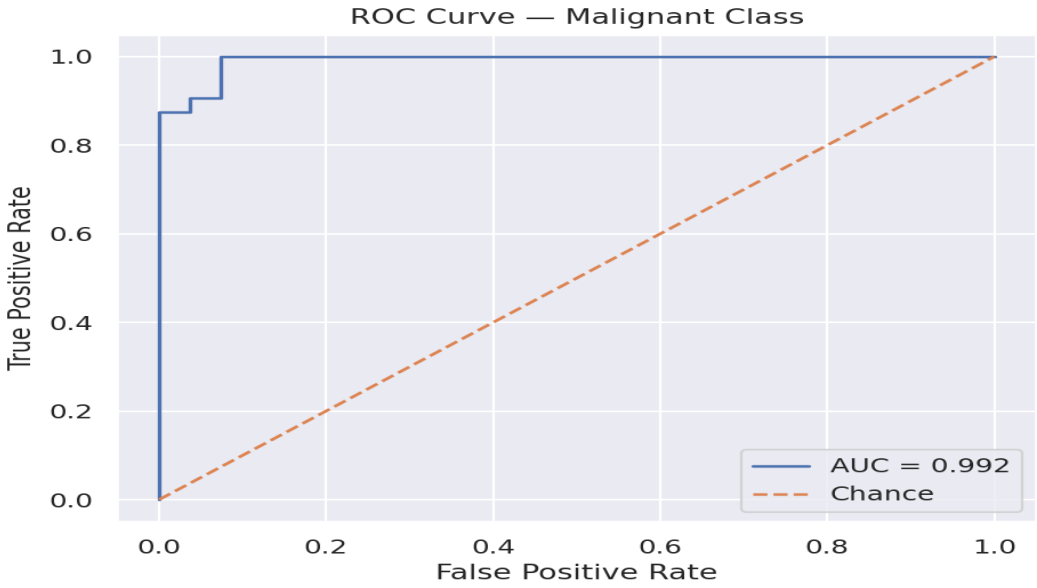
The model's internal validation score is monitored during early stopping. Fluctuations are expected on a small validation subset; the score is not the same as the separate held-out validation partition accuracy.

Figure 8. Confusion matrix on the test set



Rows represent actual classes and columns predicted classes. Off-diagonal cells are mistakes. In a medical-data demonstration, false negatives and false positives have different implications and should be examined separately.

Figure 9. ROC curve



The ROC curve plots true-positive rate against false-positive rate across thresholds. The reported AUC summarizes ranking discrimination for the malignant class on this test split; a high AUC does not guarantee safe clinical use.

7. Statistical Interpretation

Univariate analysis describes individual feature distributions. The class distribution shows more benign than malignant records, supporting use of class-sensitive metrics rather than accuracy alone. The boxplot summarizes median and spread and flags potential outliers for investigation. The correlation heatmap reveals redundancy among size-related variables, while the scatterplot shows whether two measurements jointly separate target classes.

The assignment asks for discrete/continuous probability distributions wherever applicable. Here, input measurements are continuous and the target is categorical. Rather than forcing a Poisson or binomial model onto arbitrary feature columns, a defensible extension would test whether selected continuous features are approximately normal using Q–Q plots and suitable goodness-of-fit checks. The binary target can be described using class proportions. Distributional claims should be supported by assumptions and tests.

8. Conclusion and Future Scope

This case study demonstrates an EDA-to-modeling workflow for a 569-row diagnostic dataset with 30 input features. The included CSV had no missing values or exact duplicate rows. Five EDA visualizations examined class balance, a feature distribution, group spread, correlations and a bivariate relationship. The MLP achieved 94.2% test accuracy and 0.992 ROC-AUC for the malignant class on the fixed split. These results do not guarantee performance on unseen clinical populations.

Limitations: small dataset, potential sampling bias, one fixed split, sensitivity to random seed and hyperparameters, and no external clinical validation. High scores do not imply clinical readiness.

Future scope: repeated stratified cross-validation, systematic hyperparameter tuning, comparison with logistic regression and tree-based baselines, calibration and threshold analysis, confidence intervals, independent-dataset evaluation and domain review before any real-world medical use.

9. References

[1] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[2] Scikit-learn developers, “Breast cancer wisconsin (diagnostic) dataset.” https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

[3] UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic).” https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

[4] Scikit-learn developers, “MLPClassifier.” https://scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPClassifier.html

[5] Scikit-learn developers, “StandardScaler.” https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html

10. Appendix: Execution Instructions

1. Install Python 3.10 or later.
2. Extract the ZIP and open a terminal in the project folder.
3. Create and activate a virtual environment.
4. Run `pip install -r requirements.txt`.
5. Run `python src/run_analysis.py`.
6. Review generated charts in `visualizations/` and metrics in `report/metrics.json`.

Before submission, fill in your name and roll number, review the report, run the code yourself, and ensure you can explain the dataset, preprocessing, model and evaluation. Create a public GitHub repository under your own account and submit its link as required.

Additional Statistical Analysis: Probability Distributions

This section supplements the distribution discussion in the main report and directly addresses the assignment rubric for discrete/continuous probability distributions wherever applicable.

Continuous feature: mean radius

The mean-radius feature has n = 569, mean = 14.127, median = 13.370, standard deviation = 3.524, skewness = 0.942, and excess kurtosis = 0.846. The mean is greater than the median and the skewness is positive, consistent with a right-skewed distribution.

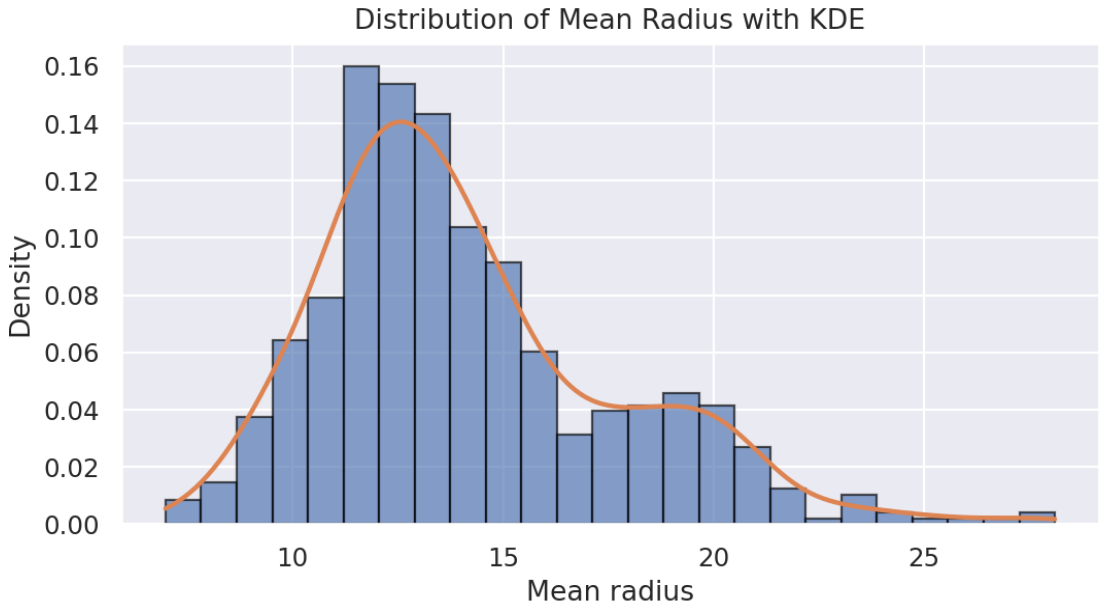


Figure 10. Histogram and kernel density estimate for mean radius. The histogram displays observed frequency density; the KDE is a smoothed estimate of the sample distribution, not proof that the data follows a named theoretical distribution.

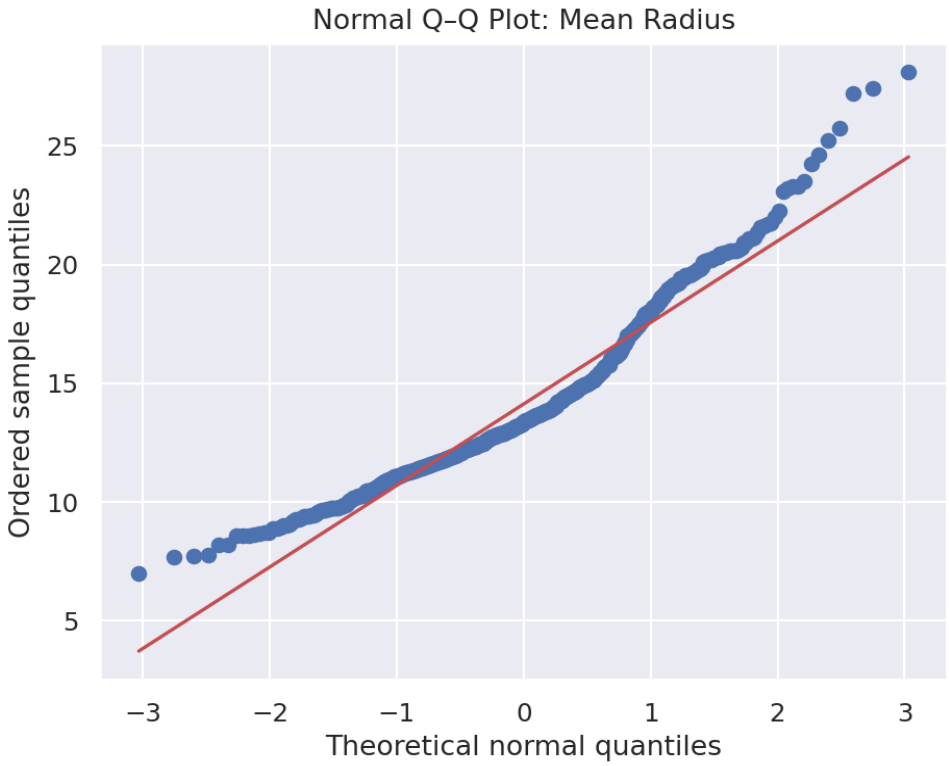


Figure 11. Normal Q–Q plot for mean radius. Departures from a straight reference line indicate deviations from a normal distribution, particularly in the tails.

Shapiro–Wilk normality test: W = 0.9411, p = 3.11e-14. At alpha = 0.05, reject the null hypothesis that this sample is drawn from a normal distribution. The p-value is evidence against normality; it does not identify a unique alternative distribution and should be interpreted alongside the plots and sample size.

Discrete/categorical target distribution

The target is binary rather than a continuous measurement. In this dataset, 212/569 (37.3%) observations are malignant and 357/569 (62.7%) are benign. These are empirical class proportions. A binomial model may be relevant to counts under an appropriate sampling model, but the target labels should not be treated as a Poisson event-count variable. The project therefore describes the observed categorical distribution rather than forcing an unsuitable probability model.

Interpretation and limitations

The continuous feature is not well described by a normal distribution in this sample. This matters because analyses that assume normality should justify that assumption rather than presume it. Distributional analysis is descriptive and does not establish causal relationships or clinical validity.